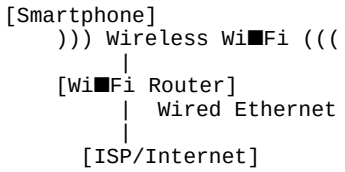
Wi-Fi Networking Assignment Report

1. Popular Wi■Fi Standards

802.11n (Wi■Fi 4): Up to ~600 Mbps, operates on 2.4 GHz and 5 GHz.
802.11ac (Wi■Fi 5): Higher speeds than 802.11n, operates mainly on 5 GHz.
802.11ax (Wi■Fi 6): Improved efficiency and higher throughput, supports 2.4 GHz and 5 GHz.

2. Home Wi■Fi Diagram



3. Wi■Fi Network Scan (Sample Example)

SSID	Signal Strength	Security
HomeNet	-45 dBm	WPA3
NeighborWiFi	-60 dBm	WPA2
Cafe_FreeWiFi	-75 dBm	Open

Preferred Network: HomeNet because it has the strongest signal and uses WPA3 security, providing better protection and performance.

4. SSID Hiding

SSID hiding means a wireless network does not broadcast its network name publicly. Advantage: casual users cannot immediately see the network in available Wi■Fi lists. Disadvantage: it does not provide strong security because attackers can still discover the SSID using network analysis tools, and legitimate users must manually enter the network name.